

Integrated Waste Management and Environmental

Integrated Waste Management and Environmental
Sustainability System for LGU MINA

A Capstone Project

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College of Information Technology Education
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In Partial Fulfillment
of the Requirements for the degree
Bachelor of Science in Information Technology

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled Integrated Waste Management and Environmental Sustainability System for LGU MINA was prepared and submitted by Flores, Sean Kristopher Jumilla, Adrian Salanio, Geran Lister, and Valdez, Felice Jude, who are hereby recommended for the corresponding final evaluation.

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Abstract

This capstone project developed an Integrated Waste Management and Environmental Sustainability System for LGU Mina to address persistent issues such as improper waste segregation, low recycling rates, and limited community participation in environmental programs. The system enables residents to log their waste by category, earn digital reward points for proper segregation, and access educational resources to enhance their knowledge and practices on waste management.

LGU staff can validate submissions, manage user rewards, and generate compliance and performance reports through an analytics dashboard for better policy-making and monitoring. Using the Software Development Life Cycle (SDLC) model, the system underwent systematic stages including planning, analysis, design, development, testing, and pilot implementation in selected barangays.

Results showed improved resident awareness, motivation to segregate waste properly, and overall satisfaction with the digital platform. This suggests the system's potential to support effective environmental governance, promote sustainable waste management practices, and serve as a replicable model for other municipalities aiming to integrate technology with their environmental programs.

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Chapter I

Introduction to the Study

The increasing concerns about waste management and environmental sustainability call for the application of the innovative and integrated approaches, above all, for the Local Government Units (LGUs) in the Philippines. In this context, the capstone project presents the idea of "Integrated Waste Management and Environmental Sustainability System for LGU Mina" offering the blend of ideas like "EcoMina" website intended to change the whole scenario in waste management, continuity of environmental stewardship decision-making processes, and community involvement.

The EcoMina website, on the one hand, invites all people to turn the recyclable waste into transport credits thus, on the other hand, promotes the direct involvement of the community in the recycling programs. The system, which is made up of sensor networks, advanced data analytics, and a user-friendly web system, will provide LGU Mina with the tools and insights required to optimize waste collection, reduce environmental impact, and achieve a more sustainable future.

With the use of this technology, waste collection will be faster, data will be monitored in real time, public participation will be increased, and through automation and analytics, the local government services will be able to provide a better service.

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Technical Background of the Study

Waste management in a proper way is very important for the health of the public, for the environment and for the economy as well. Often, traditional garbage handling techniques do not deal properly with the complex nature of today's waste streams. The Integrated Waste Management and Environmental Sustainability System operates with the adoption of various key technologies:

- **Data Analytics:** Process vast amounts of waste-related data to identify trends and optimize resource allocation.
- **EcoMina Website:** A web-based platform that incentivizes community participation by allowing users to: Submit recyclable waste information (plastic, metal, or paper/cardboard) to earn points.
- Track earned points and revenue from recycling activities.
- View total revenue and monetization reports.
- Redeem points for credits instantly.
- View community recycling statistics and trends.
- Stay informed about recycling programs and events.

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Statement of the Problem

Mina, the waste management of Iloilo goes through the problem of improper tracking, a lack of motivation, and underdeveloped infrastructure. This research explores the EcoMina system with the purpose of finding out the following:

1. How can digital records be used by EcoMina to track the collection and compliance of the residents effectively?
2. What are the digital tools that can best replace waste logging, point tracking, and reporting which are manual processes?
3. In what ways can the design of the platform along with the incentives render the participation of the residents to a higher extent?
4. What are the specific features (such as real-time feedback, and analytics) that can make the platform easy to access, and useful for residents and administrators as well?

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Objectives of the Study

The main objective of this project is to develop and implement a functional, secure, web-based waste management system that promotes segregation, supports LGU environmental planning, and incentivizes sustainability among residents.

Specific Objectives:

- To design and develop a user-friendly platform that allows residents to accurately log segregated waste by type.
- To implement a transparent, points-based reward system that immediately encourages and sustains resident participation in recycling efforts.
- To develop administrative tools and an analytics dashboard for LGU personnel for monitoring, validation, and comprehensive reporting.
- To provide accurate, real-time data and metrics for evaluating waste segregation performance and compliance rates.
- To increase public awareness through integrated education modules and timely notifications regarding proper waste handling.
- To assess the system's usability, engagement level, and overall impact during the defined pilot deployment phase.

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Significance of the Study

The Municipality of Mina, in Iloilo, gets the most out of the technology-driven practical approach to waste management that this study presents. The system is to empower different sectors: Local Government Unit (LGU) of Mina: The system offers officials real-time and analytics-based data which they can use to monitor segregation compliance, resource allocation and decision-making by policy inferred from evidence.

Residents of Mina: The digital rewards system for proper waste disposal offers the residents an immediate and noticeable benefit, which in turn leads to their active and responsible involvement in the community's ecological goals.

Educational and Awareness Campaigns: The platform is a digital center for disseminating guides, alerts, and educational content that will help sustain a culture of eco-friendliness all over the place.

Recycling Partners and the Circular Economy: The system eliminates the possibility of mixing recyclable materials at the source which in turn enhances the quality and supports the sustainability of the circular economy.

Academic and Technological Sector: The research is a significant contribution to the ongoing study of digital environmental solutions and it also offers a verifiable instance of smart local governance and community engagement via digital transformation.

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Scope of the Study

The paper presents the procedure, conception, and testing of an online Integrated Waste Management and Environmental Sustainability System (EcoMina) designed for LGU Mina and implemented in a limited manner.

Waste logging by category, a digital rewards mechanism to attract residents, an admin dashboard for monitoring and validation, and educational modules to support proper waste segregation and recycling habits are the main functionalities of the system. Its main purpose is to facilitate participation of the community and the LGU by supporting their environmental sustainability through the use of digital technology.

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Limitation of the Study

While the EcoMina offers many helpful features, there are still some limitations:

1. **Geographical Limitation:** the implementation of a pilot phase is confined to the targeted barangays in the municipality of Mina; it does not extend to the whole municipality.
2. **Technical Limitation:** the project will only have a web-based application and will not develop a separate mobile application or use Internet of Things (IoT) technologies (e.g., smart bins) for integration.
3. **Data Dependency:** the correctness of the user-submitted data depends on the residents' honesty and loyalty in reporting waste entries, which might need LGU's manual validation.
4. **External Factors:** the system's usability and user engagement level might be impacted by the internet connectivity issues in some municipal areas that exist already.

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Definition of Terms

1. **Integrated Waste Management** (IWM) - A method that incorporates collection, sorting, recycling, and disposal techniques in an organized manner to effectively and responsibly manage waste.
2. **Environmental Sustainability** - The method of using natural resources wisely and keeping nature in balance to the advantage of both humans and nature; that is, the present and the future.
3. **Waste Segregation** - The separation of waste into different groups such as compostable, non-compostable, and recyclable for their proper disposal and recovery is called waste segregation.
4. **Recyclables** - Items like paper, plastic, glass, and metals that can be collected, processed, and reused thus lessening the negative impact on the environment.
5. **Reward System** - A digital incentive characteristic that supplies the residents with points or benefits in return for proper waste classification and recycling activities.
6. **Green Wallet** - An electronic wallet embedded in the system where people living in a particular area gather and control the reward points obtained through eco-friendly practices.

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7. **Analytics Dashboard** - An interface that displays graphs, measures, and reports which enable local government unit (LGU) officials to keep track of segregation rates, recycling volume, and resident participation.
8. **Local Government Unit (LGU)** - The administration that is in charge of executing waste management policies and programs within the community, including environmental management initiatives, among others.

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Chapter II

Literature Review

Integrated waste management (IWM) is a crucial strategy for promoting environmental sustainability at the local government unit (LGU) level, and in the context of Mina, Iloilo this becomes particularly relevant given its rural character and agricultural base. The legal foundation for IWM in the Philippines is provided by Republic Act No. 9003 Ecological Solid Waste Management Act of 2000, which mandates LGUs to implement segregation at source, establish barangay level Material Recovery Facilities (MRFs), prepare ecological solid waste management plans, and promote resource recovery rather than mere disposal (Republic Act No. 9003, 2000). This law remains the anchor for Mina's efforts to build a stronger waste management system. Despite the strong legal framework, many LGUs especially smaller, predominantly agricultural municipalities like Mina face implementation gaps due to limited technical capacity, financial constraints, and sometimes weak monitoring or enforcement structures (Guerrero, Maas & Hogland, 2013).

In Mina's case, the municipality's own planning documents reveal key characteristics that both shape and challenge its waste management system. According to its "10 Year Ecological Solid Waste Management Plan (2015-2024)", Mina has a land area of about 4,340 hectares and is located approximately 38 km north of Iloilo City. The plan further reveals that the majority of waste generated in Mina is from the household level with about 7,200 kg collected daily, a yearly equivalent of approximately 2,625.99 tons from households alone (Comprehensive Land Use

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Plan, Mina). In terms of composition, about 61 % of the municipality's solid waste is biodegradable/compostables (e.g., food waste, garden waste, twigs, leaves) and around 27 % is recyclable (e.g., paper, cardboard, glass, tin cans). These data underscore that for Mina, the bulk of the waste stream is organic and thus presents strong opportunities for composting and resource recovery approaches, a key advantage if the IWM system is designed for circular economy linkages (e.g., organic compost for local agriculture) rather than purely disposal. At the same time, the large share of compostables signals a need for immediate infrastructure (composting systems, decentralized treatment) and community engagement (segregation at source, collection logistics) to capture this value.

Several studies emphasise that local level waste management systems must integrate not just technical infrastructure but institutional, social and behavioural dimensions. For example, accurate waste characterization like Mina has undertaken is foundational for sizing MRFs, collection routes, composting systems, financial modelling and decision-making (Gonzales & Magcale-Macandog, 2021). Collaboration of stakeholders (LGU staff, barangays, households, informal waste sector, recyclers) is another key factor for success. This is relevant in Mina's context where barangay-level engagement will likely determine the performance of segregation at source campaigns and community based composting. Also, partnerships with private recyclers and the informal waste sector can help increase recovery rates and reduce landfill dependence (Guerrero et al., 2013). Given that Mina is largely rural and agricultural, linking waste management

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with local agriculture (e.g., compost for farms) offers a strong path toward sustainability and local value creation.

From an environmental sustainability perspective, Mina's IWM system must do more than manage disposal; it must aim toward resource recovery, pollution prevention and circularity. The high proportion of compostable waste means that a composting programme (or anaerobic digestion) could reduce methane emissions, lower disposal costs, and produce useful soil ameliorants for local agriculture. Studies in Philippine LGUs show that integrated approaches combining segregation, composting, education and recycling achieve higher diversion and better community outcomes than those relying solely on collection and disposal (Magalang & Caballes, 2019). In Mina, which has a strong agriculture base and likely benefits from locally sourced compost, this creates an especially relevant synergy.

Despite the positive context, gaps remain in Mina's system and in wider literature. First, although Mina has data on waste volumes and composition, there is limited evidence that these data are regularly updated or used in dynamic decision making (e.g., seasonal variation, informal waste streams, e-waste or hazardous household waste). The municipality's plan covers up to 2024, and forward-looking updates may be needed. Second, while the policy framework mandates resource recovery, many LGUs struggle to embed sustainability indicators (such as carbon footprint reduction, circular economy performance, community resilience) in their SWM plans. Third, capacity constraints (budget, technical staff, operational systems) remain a hurdle

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for fifth class municipalities like Mina. Finally, as one news article indicates, social acceptance can be a critical barrier. In a past case in Mina, residents opposed a proposed sanitary landfill citing environmental and health concerns (Philippine News Agency, 2018). These are important to keep in mind in system design.

References

Guerrero, L. A., Maas, G., & Hogland, W. (2013). Solid waste management challenges for cities in developing countries. *Waste Management*, 33(1), 220-232.

Gonzales, M., & Magcale Macandog, D. (2021). Integrated solid waste management practices in Philippine municipalities: Challenges and prospects. *Journal of Environmental Science and Management*, 24(2), 56-68.

Magalang, A., & Caballes, M. (2019). Linking waste management and climate action in Philippine LGUs. *Philippine Journal of Environmental Sustainability*, 4(1), 45-59.

Mina Municipal Government. (2015). 10 Year Ecological Solid Waste Management Plan (2015-2024). Municipality of Mina, Iloilo.

Municipality of Mina. (2016). Comprehensive Land Use Plan, Volume 1 (2016-2025). Mina, Iloilo.

Philippine News Agency. (2018, May 10). Residents oppose proposed sanitary landfill in Iloilo town.

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Chapter III

3.1 Methodology

This chapter provides a detailed explanation of the approach used in the design, development, and implementation of the Integrated Waste Management and Environmental Sustainability System for LGU Mina, which includes the EcoMina website. It thoroughly describes every aspect of the process, from the preliminary research and development to the final deployment and maintenance phases.

The chapter discusses the research and development stage, the Software Development Life Cycle (SDLC), data collection techniques, the justification for using the Agile methodology, system architecture and design (including detailed explanations of the ERD and DFD), database design, user interface design, security considerations, technology stack, data gathering and analysis techniques, and ethical considerations.

3.2. Research and Development Stage

The research and development stage was a fundamental phase that involved a thorough assessment of current waste disposal systems, technologies, and standard procedures. This involved conducting extensive research on community participation tactics and incentive programs in order to drive the design and development of LGU Mina's Integrated Waste Management and

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Environmental Sustainability System. This stage included the following activities:

Literature Review: A thorough examination of educational papers, reports from industries, government documents, online databases, and any other relevant sources to discover applicable ideas, procedures, innovations, and best practices. The literature review centered on the following main areas:

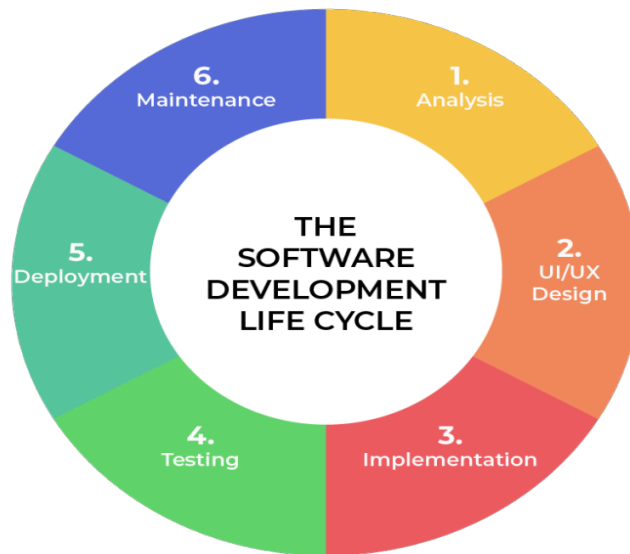
- **Analytics of Data for Waste Management:** Investigating data mining techniques, learning methods, and statistical approaches for analyzing waste data, forecasting trash creation rates, and assessing the success of waste management programs.
- **Community Engagement Strategies in Waste Management:** Investigating various methods for including communities in waste management activities, such as public awareness campaigns, educational programs, and incentive-based schemes.
- **Recycling and Waste Reduction Incentive Programs:** Investigating various sorts of incentives (e.g., monetary awards, discounts, and interactive features) for encouraging recycling and waste decrease, as well as their efficiency and cost-benefit analysis.
- **Interviews:** Formal and somewhat structured conversations with important stakeholders were conducted to acquire full

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details about the present waste management system, its benefits and flaws, the issues encountered by LGU Mina, and the anticipated objectives of the proposed system.

3.3. Software Development Life Cycle (SDLC)



The Software Development Life Cycle (SDLC) provided a structured and systematic approach to the design, development, testing, and deployment of the Integrated Waste Management and Environmental Sustainability System for LGU Mina, including the EcoMina website. The SDLC phases included:

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Planning: Defining the project goals, scope, objectives, deliverables, timelines, and resources. This involved:

- Developing a detailed project management plan outlining the tasks, responsibilities, and milestones for each phase of the SDLC.
- Creating a work breakdown structure (WBS) to decompose the project into smaller, manageable tasks.
- Estimating the time and resources required for each task.
- Identifying potential risks and developing mitigation strategies.
- Establishing communication protocols and reporting mechanisms.

Analysis: Gathering and analyzing requirements from stakeholders to understand their needs, expectations, and priorities. This involved:

- Conducting interviews with waste management personnel, local officials, community members, and EcoMina website users to elicit detailed requirements for the system and the website.
- Developing use cases to describe how users will interact with the system and the website.
- Creating user stories to capture the requirements from the user's perspective.
- Developing wireframes and mockups to visualize the user interface and user experience.
- Prioritizing requirements based on their importance and feasibility.

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Design: Creating the system architecture, database design, user interface design, and security design based on the requirements gathered in the analysis phase. This involved:

- Developing a detailed system architecture diagram to illustrate the components, modules, and interfaces of the system.
- Creating an entity-relationship diagram (ERD) to model the database schema and relationships between entities.
- Designing the user interface for the system and the EcoMina website, with a focus on usability, accessibility, and aesthetics.
- Developing a security design to protect the system and the data from unauthorized access and cyber threats.

Implementation: Developing and coding the system components and the EcoMina website based on the design specifications. This involved:

- Writing code in programming languages such as, JavaScript, PHP, and others.
- Implementing the database schema and data access layer.
- Integrating the different components and modules of the system.
- Conducting code reviews to ensure code quality and adherence to coding standards.

Testing: Evaluating the system's functionality, performance, security, and usability to ensure it meets the specified requirements and performs as intended. This involved:

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- Conducting unit tests to verify the functionality of individual components and modules.
- Conducting integration tests to verify the interaction between different components and modules.
- Conducting system tests to verify the overall functionality of the system.
- Conducting user acceptance tests to verify that the system meets the needs of the users.
- Conducting security tests to identify and address potential security vulnerabilities.
- Conducting performance tests to evaluate the system's performance under different load conditions.

Deployment: Installing and launching the system and the EcoMina website in LGU Mina. This involved:

- Setting up the hardware, software, and network infrastructure.
- Deploying the system and the website to a production environment.
- Configuring the system and the website to meet the specific needs of LGU Mina.
- Training waste management personnel and community members on how to use the system and the website.

Maintenance: Providing ongoing support, updates, and enhancements to the system and the EcoMina website to ensure continued operation and effectiveness. This involved:

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- Monitoring system performance and identifying potential issues.
- Responding to user feedback and addressing any problems or concerns.
- Implementing new features and functionalities based on user feedback and changing requirements.
- Applying security patches and updates to protect the system from cyber threats.
- Performing regular backups to ensure data integrity and availability.

3.4. Data Collection Techniques

Data collection techniques were used to gather comprehensive and reliable data about waste generation, collection, and disposal in LGU Mina, as well as community attitudes and participation in recycling. These techniques included:

Surveys: Administering questionnaires to residents, businesses, and institutions to gather quantitative and qualitative data on:

- Waste generation habits (e.g., types and quantities of waste generated, frequency of waste disposal)
- Recycling practices (e.g., types of materials recycled, participation in recycling programs)
- Attitudes towards waste management (e.g., awareness of waste management issues, willingness to participate in waste reduction efforts)
- Awareness of and interest in the EcoMina program (e.g., knowledge of the program, likelihood of participating)

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- Preferences for rewards (mobile load credits) (e.g., preferred denominations, usage patterns)
- Demographic information (e.g., age, gender, education level, income)
- The surveys were administered using a combination of online and paper-based methods to reach a wide range of respondents.
- **Observations:** Conducting direct observations of waste collection routes, waste disposal sites, and recycling facilities to assess:
 - The efficiency of waste collection routes (e.g., time taken to complete routes, fuel consumption)
 - The effectiveness of waste disposal sites (e.g., capacity, environmental impact)
 - The effectiveness of recycling facilities (e.g., processing capacity, material recovery rates)
 - The presence of illegal dumping and littering
 - The behavior of waste management personnel and community members
 - The observations were documented using checklists, photographs, and videos.

Measurements: Measuring waste volumes and weights at collection points, transfer stations, and disposal sites to quantify:

- Waste generation rates (e.g., kilograms of waste generated per household per day)
- Waste composition (e.g., percentage of different types of materials in the waste stream)

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- Waste disposal patterns (e.g., percentage of waste sent to landfills, recycled, or composted)
- The measurements were conducted using calibrated scales, volume meters, and GPS devices.

Document Analysis: Reviewing existing waste management plans, reports, regulations, and policies to understand:

- The legal and institutional framework for waste management in LGU Mina
- The goals and objectives of the waste management system
- The roles and responsibilities of different stakeholders
- The performance of the waste management system
- The financial resources allocated to waste management
- The document analysis also included a review of relevant environmental data, such as air quality, water quality, and soil contamination reports.

Website Analytics: Tracking user activity on the EcoMina website using tools like Google Analytics to gather data on:

- Website traffic (e.g., number of visitors, page views, session duration)
- User demographics (e.g., age, gender, location)
- User behavior (e.g., pages visited, time spent on each page, navigation patterns)
- Submission rates (e.g., number of waste submissions per user, types of waste submitted)

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- Rewards redemption rates (e.g., number of rewards redeemed per user, types of rewards redeemed)
- Conversion rates (e.g., percentage of visitors who register for the EcoMina program)
- The website analytics data was used to evaluate the effectiveness of the EcoMina website and identify areas for improvement.

3.5. Justification for Selecting the SDLC Methodology

The Systems Development Life Cycle (SDLC) methodology was selected for the development of this web-based system because it provides a clear, organized, and disciplined approach to software creation. Since the system involves multiple modules such as user authentication, dashboard management, categorization, and reward claiming, SDLC ensures that each component is carefully analyzed, designed, tested, and implemented according to user and business requirements. This structured process allows developers to maintain consistency, reduce system errors, and ensure the project meets its objectives before deployment.

ASPECTS | SDLC METHOD | AGILE APPROACH

ASPECTS	SDLC METHOD	AGILE APPROACH
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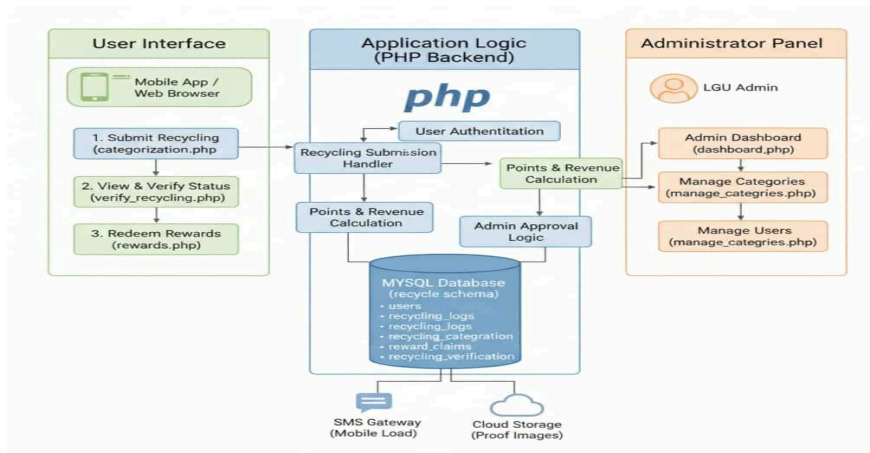
Development Style	Follows a sequential process: planning, analysis, design, implementation, testing, deployment, and maintenance – ensuring that modules like login, dashboard, and rewards are developed in order.	Iterative and incremental; delivers partial features after short development sprints.
Flexibility	Limited flexibility once the process begins; major changes require revisiting previous stages.	Highly flexible; allows frequent changes and improvements throughout the project.
User Involvement	Users participate mainly during the requirement and testing stages to confirm that the system meets their needs.	Users are actively involved at every stage, providing continuous feedback.
Delivery	The complete and stable system is delivered after thorough testing and evaluation.	Partial system features are released continuously after each sprint.
Risk Management	Risks are identified early during planning and documentation, reducing the	Risks are managed continuously through short

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	chance of critical issues during implementation.	iterations and frequent reviews.
Testing	Conducted after the development phase to ensure the accuracy of modules such as login validation, data management, and reward tracking.	Continuous testing is done after each sprint for new updates.
Collaboration	Involves formal communication between developers and users mainly during review and approval phases.	Promotes daily collaboration and direct communication between clients and developers.
Adaptability to Change	Moderate – changes can be made, but require re-documentation and adjustment of earlier phases.	Excellent – quickly adapts to new requirements and feedback.
Efficiency	Ensures a well-structured, reliable, and secure system through complete documentation and systematic execution.	Promotes faster development but may lead to inconsistencies without proper documentation.

3.6. System Architecture and Design



3.6.1. System Architecture

The system architecture provides a comprehensive overview of the structural organization of the Integrated Waste Management and Environmental Sustainability System, including the EcoMina website. It delineates the key components, modules, interfaces, and data flows that constitute the system. The system architecture is meticulously designed to be scalable, modular, and interoperable, ensuring its ability to accommodate future growth and seamless integration with other systems. The system architecture comprises the following distinct layers:

- Communication Layer:** The communication layer acts as the intermediary, providing the communication protocols, networks, and gateways that facilitate the secure and reliable transmission of data from the sensor layer to the data processing layer.

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- **Data Processing Layer:** The data processing layer serves as the analytical engine, housing the data storage, data analysis, and data management components that process the raw data collected from the sensor layer, transforming it into actionable insights.

3.6.2. System Components and Modules

The system components and modules constitute the building blocks of the Integrated Waste Management and Environmental Sustainability System. Each component and module is meticulously designed to perform a specific function and can be seamlessly integrated with other components and modules to create a cohesive and comprehensive system. The system components and modules include:

- **Data Analysis Module:** Employs sophisticated data analysis techniques to identify trends, predict future waste generation rates, and evaluate the effectiveness of waste management programs.
- **Reporting Module:** Generates comprehensive reports on waste generation, collection, and disposal, providing valuable insights for decision-making.
- **User Management Module:** Manages user accounts, defines access permissions, and assigns roles to ensure system security and data integrity.

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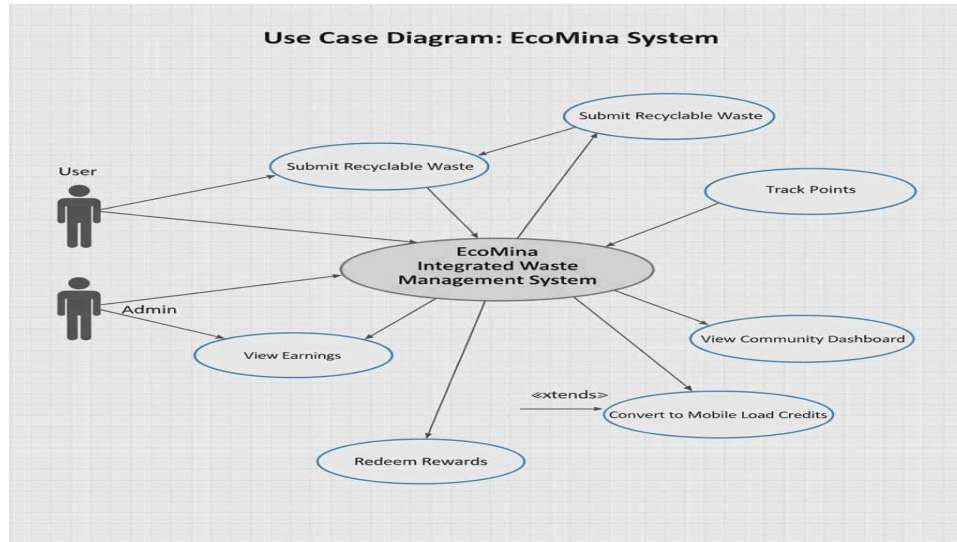
EcoMina Website Modules:

- **Submission Module:** Enables users to categorize and submit information about recyclable waste (plastic, metal, or paper/cardboard) to earn points, fostering community participation in recycling efforts.
- **Points Tracking Module:** Provides users with a transparent view of their earned points and revenue generated from their recycling activities, incentivizing continued participation.
- **Earnings Viewing Module:** Allows users to access detailed reports on their total revenue and monetization, providing a clear understanding of the economic benefits of recycling.
- **Rewards Redemption Module:** Empowers users to convert their accumulated points into tangible rewards, such as mobile load credits, creating a direct and immediate incentive for recycling.
- **Analytics Dashboard Module:** Presents a user-friendly dashboard that visualizes community recycling statistics and trends, promoting awareness and encouraging further participation.
- **Community Updates Module:** Keeps users informed about upcoming recycling programs, community events, and other relevant information, fostering a sense of community and shared responsibility.

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3.6.3. Use Case Diagram

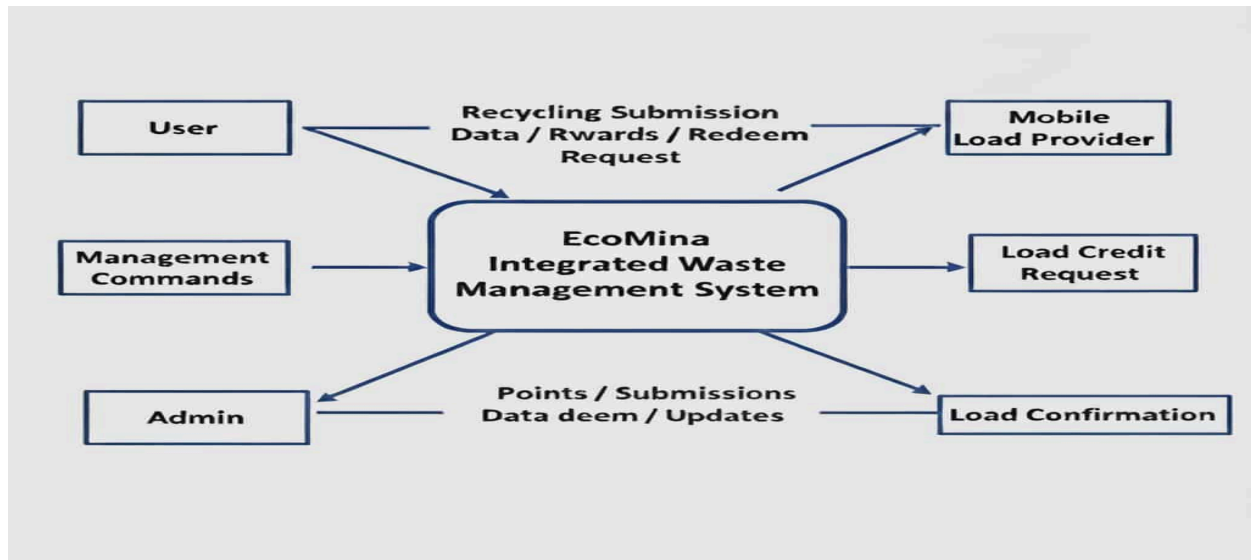


3.6.4. Data Flow Diagram

This diagram highlights the flow of data from waste collection points to the central database, the processing of that data for analysis and reporting, and the interaction of users with the EcoMina website to submit waste information and redeem rewards.

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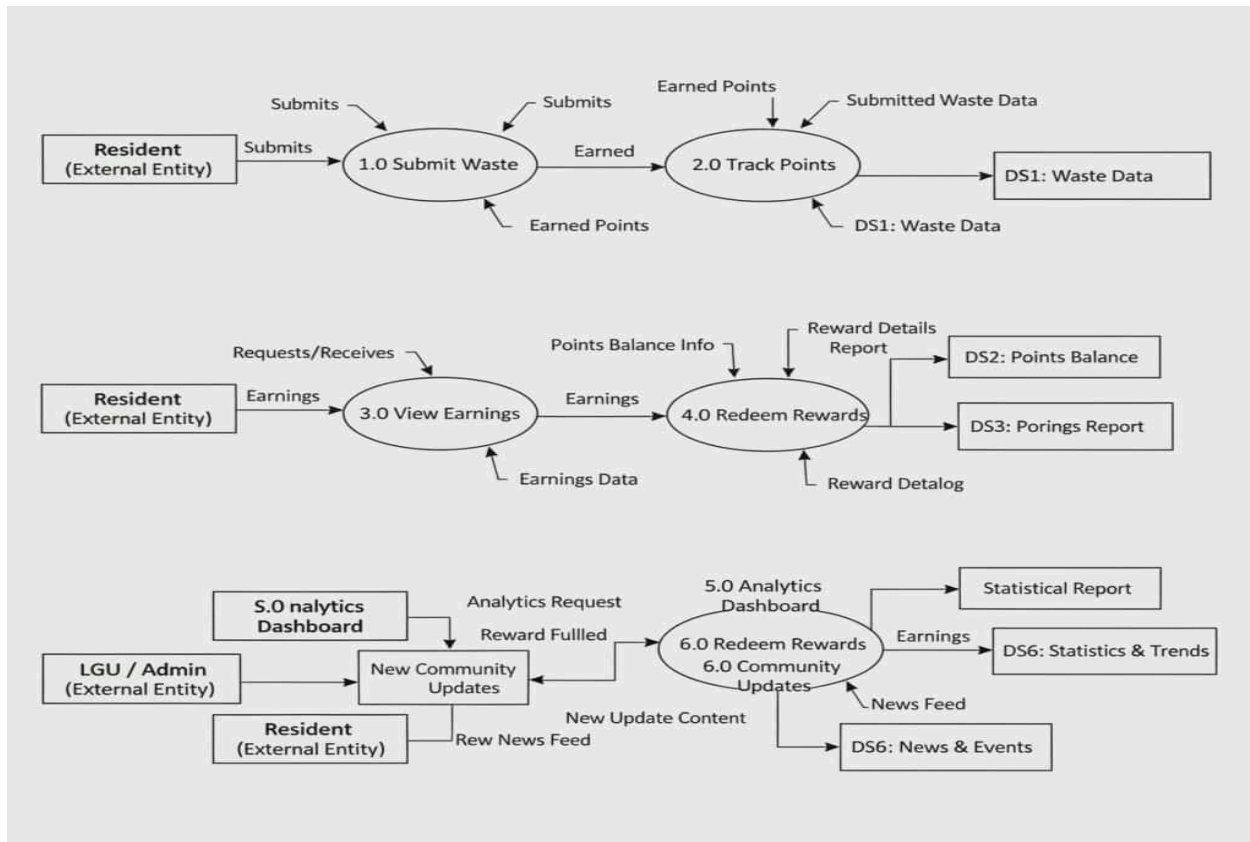
Level 0 Data Flow Diagram

The diagram shows the EcoMina Integrated Waste Management System as the central process, interacting with external entities:

- **User:** Submits Recycling Data and initiates Reward/Redeem Requests to EcoMina.
- **Mobile Load Provider:** Receives Load Credit Requests from EcoMina, likely to fulfill user rewards.
- **Load Confirmation:** Provides Load Confirmation back to EcoMina after a reward is processed.
- **Admin:** Manages the system by providing Points/Submissions Data, Redeem/Updates to EcoMina.
- **Management:** Provides Management Commands to the EcoMina system.

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Level 1 Data Flow Diagram

This simplified DFD will showcase the following key data flows:

- **Data Input:** Waste data from collection points and user submissions via the EcoMina website.
- **Data Processing:** Data analysis, validation, and storage in the central database.
- **Data Output:** Reports, analytics, and reward calculations for EcoMina users.

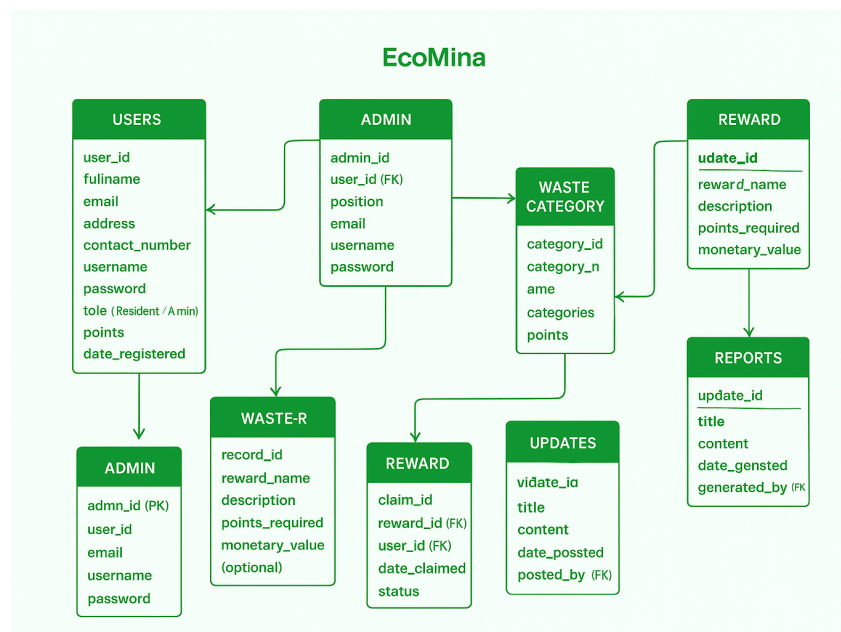
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- **External Entities:** Waste collection personnel, LGU officials, and EcoMina website users.

3.6.5. ERD (Entity Relationship Diagram)

Figure 3.2 shows a simplified Entity Relationship Diagram illustrating the main entities and their relationships in the system database, including entities related to the EcoMina website. This diagram highlights the relationships between users, waste types, collection points, rewards, and other key data elements. A more detailed ERD can be found in Appendix B.



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Database Design

3.8. User Interface Design

The user interface design is a critical aspect of the project, focusing on the creation of user-friendly and intuitive interfaces for all stakeholders, including waste management personnel, community members, and LGU officials. The design principles prioritize usability, accessibility, and aesthetics to ensure a positive user experience and maximize system adoption. This includes the design of:

- **Dashboards:** Customizable dashboards to provide users with a comprehensive overview of key metrics and performance indicators.
- **Forms:** User-friendly forms for data entry, submission, and management.
- **Reports:** Clear and concise reports for data analysis and decision-making.
- EcoMina Website User Interface:
- Clean and intuitive layout to facilitate easy navigation and information access.
- Visually appealing design to enhance user engagement and create a positive experience.
- Clear instructions and feedback to guide users through the process of submitting waste and redeeming rewards.
- Gamified elements (e.g., points, badges, leaderboards) to incentivize participation and make the recycling process more engaging.

3.9. Security Considerations

Security considerations are paramount in the design of the Integrated Waste Management and Environmental Sustainability System, with a strong emphasis on protecting the confidentiality, integrity, and availability of the data. The security design includes a multi-layered approach to mitigate potential risks and ensure data security. This includes:

User Authentication and Authorization: Secure user authentication mechanisms (e.g., username/password, multi-factor authentication) to verify user identities and prevent unauthorized access. Role-based access control to restrict user access to only the data and functionalities they are authorized to use.

- **Data Encryption:** Encryption of sensitive data both in transit (using HTTPS) and at rest (using encryption algorithms) to protect against data breaches.
- **Access Controls:** Strict access controls to limit physical and logical access to the system and its components.
- **Firewall Protection:** Implementation of firewalls to protect the system from unauthorized network access and malicious attacks
- **Intrusion Detection:** Intrusion detection systems to monitor network traffic and system activity for suspicious behavior.
- **Security Audits:** Regular security audits to identify and address potential vulnerabilities in the system.

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EcoMina Website Security:

- Secure user registration and login processes to protect user accounts and prevent unauthorized access.
- Protection against common web vulnerabilities (e.g., SQL injection, cross-site scripting) to prevent malicious attacks.
- Secure storage of user data, including personal information and points balances.
- Regular security updates and patches to address any newly discovered vulnerabilities.

3.10. Technology Stack

The technology stack encompasses the hardware, software, and network infrastructure employed to build and deploy the Integrated Waste Management and Environmental Sustainability System, including the EcoMina website. The selection of technologies is guided by factors such as scalability, reliability, security, cost-effectiveness, and compatibility with existing systems. The technology stack includes:

- **Programming Languages:** Php (for data analysis and backend development), JavaScript (for frontend development), PHP (for EcoMina website).
- **Database:** MySQL (for relational data storage)
- **Web Server:** Apache for hosting the EcoMina website.
- **Operating System:** Windows Server for hosting the system components.

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EcoMina Website Technologies:

- HTML, CSS, JavaScript for frontend development.
- PHP for backend development and server-side logic.
- Database: MySQL for storing user data, points balances, and other website-related information

Data Gathering Techniques

Survey Questionnaire For System Assessment

The purpose of this survey was to obtain user feedback on "LGU MINA." The information gathered was used to determine how the system will be improved based on the user feedback.

Demographic Information

1. What is your role in waste management?

- LGU staff
- Barangay Officials
- Resident
- Other (Please specify): _____

2. How long have you been involved in community waste programs?

- Less than 1 year
- 1-3 years
- 4-6 years
- More than 6 years

3. What is your area or barangay? _____

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Current Waste Management Practices

4. How is waste collection currently managed?

- Manual schedule (printed notice)
- Excel or spreadsheet tracking
- No formal system
- Other (Please specify): _____

5. How satisfied are you with the current system?

- Very Satisfied
- Satisfied
- Neutral
- Unsatisfied
- Very Unsatisfied

6. What common problems do you face?

- Missing Collection
- Overflowing bins
- Lacks of update
- No awareness
- Other (Please specify): _____

Expectations for the Proposed System

7. What features would you find very useful for you?

- Community Updates
- Analytics Dashboards
- Submit Recycling Waste
- User accounts

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- Redeem Rewards
- Other (Please specify): _____

8. How is recycling important?

- Very Important
- Important
- Neutral
- Not Important

Concerns and Readiness for Adoption

9. What concerns do you have? (Select all that apply)

- Cost
- Internet Access
- Training Needs
- Resistance to change
- Other (Please specify): _____

10. How willing are you to use a new digital system?

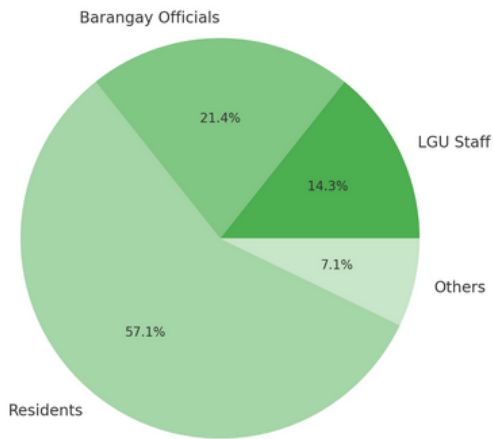
- Very Willing
- Willing
- Neutral
- Unwilling
- Very Unwilling

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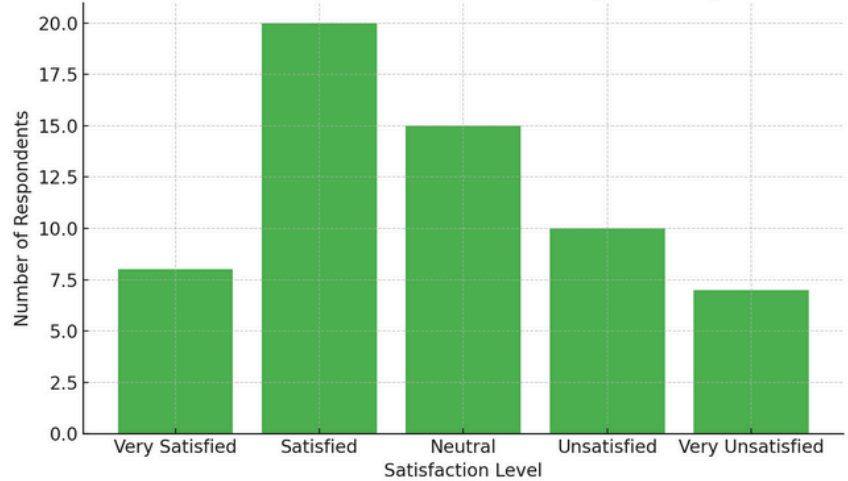
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Sampling Methods for Respondents

Respondents by Role in Waste Management (EcoMina System)



Satisfaction with Current Waste Management System



Data Analysis Techniques

The techniques used in data analysis for the Integrated Waste Management and Environmental Sustainability System for LGU Mina focus on data organization, evaluation, and interpretation concerning waste collection, categorization, and performance towards the environment. Descriptive analysis was performed to summarize data gathered from various modules such as waste volume reports, schedules for collection, and user submissions. Through this technique, administrators will easily trace the volume, type of waste generated within a specific area, and frequency of collected wastes. Prior to analysis, data validation and cleaning were performed to assure accuracy and reliability of the information stored in the database by checking for duplicate entries, incomplete records, and inconsistencies in the waste classification.

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Waste management performance was also compared for different time periods, barangays, or waste categories through comparative analysis. In this way, the LGU can identify which areas are more compliant with waste segregation and which need further improvement. Trend analysis was applied to monitor patterns over time, such as an increase or decrease in recyclable wastes, community participation, and collection efficiency. Data visualization tools involve tables, charts, and summary reports that can present the most complex data in an understandable format on the system dashboard. Different SQL-based queries were executed to retrieve, group, and filter data stored in this database and develop meaningful insights with overall performance indicators.

These data analysis techniques will generally help LGU Mina turn raw information on waste management into valuable insights to inspire better decision-making, enhance operational planning, and foster sustainable waste management behavior within the community.

3.13.Ethical Conversation

Ethical considerations were carefully addressed throughout the project to ensure responsible and ethical data handling, community engagement, and system implementation. Key ethical considerations included:

- **Data Privacy and Security:** Protecting the privacy of individuals and ensuring the security of their personal data.

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- **Environmental Impact:** Minimizing the environmental impact of the system and promoting sustainable practices.
- **Community Engagement:** Engaging the community in a meaningful and transparent way.
- **Transparency and Accountability:** Being transparent about the system's goals, methods, and outcomes, and being accountable for its performance.
- **Fairness and Equity:** Ensuring that the system is fair and equitable to all members of the community.
- **Accessibility and Inclusivity:** Making the system accessible and inclusive to all members of the community, regardless of their abilities or backgrounds.
- **Informed Consent:** Obtaining informed consent from individuals before collecting or using their personal data.

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Chapter IV

Results and Interpretation

The results indicated that the Integrated Waste Management and Environmental Sustainability System for LGU Mina effectively responded to the challenges in monitoring and managing the waste collection within the municipality. Post-deployment and evaluation of the system, data gathered from the respondents, which included administrators, waste management personnel, and selected residents, indicated satisfaction with the overall functionality, ease of use, and efficiency of the system. Respondents believed that it enhanced accuracy in tracking wastes and provided a more organized method of categorizing and recording information on wastes.

These visual reports and summaries, generated through the admin dashboard of the system, succeeded in monitoring the amount and types of waste the LGU collected within each barangay. Such reports will enable the administrators to outline trends, such as areas that have an increasing amount of recyclable materials or their irregular schedules of collection. The ability to visualize this data through charts and tables made it easier to interpret patterns, allowing the LGU to plan better waste management strategies and allocate resources more effectively.

Results from the survey and testing also showed that the system increased transparency and accountability from the level of waste collectors to the administrators. Human error was minimized as the data entry and retrieval were automated, while

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data accuracy improved remarkably over manually kept records. According to the users, categorization and reporting modules had eased their efforts with minimal paperwork, thus quicker and regular updates were made possible.

Overall, the findings analysis and interpretation establish that the system contributed positively toward the promotion of environmental sustainability and operational efficiency within LGU Mina. Digital tools and real-time data access have helped transform the manual process of waste management into a data-driven approach in support of the LGU goal of building a cleaner and more sustainable community.

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Task Manager

Week	Task	Programmer	UI/UX	Documenter
Week 1	Project Planning & Requirements	✓	✓	✓
Week 2	Research & System Design Planning	✓	✓	✓
Week 3	Use Case & DFD Creation	✓	✓	
Week 4	Database Design & schema creation	✓		
Week 5	UI Wireframes & Mock-ups	✓	✓	
Week 6	Frontend Development Begins	✓		
Week 7	7.Backend Development begins	✓		
Week 8	Integration of frontend and backend	✓		

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Week 9	System testing & debugging	✓	✓	✓
Week 10	User testing & feedback collection		✓	✓
Week 11	Final revisions (code + ui + content)	✓		
Week 12	Documentation finalization & formatting	✓	✓	✓

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Presentation of results

Impact of IWMESS on the Information Technology Field

The Integrated Waste Management and Environmental Sustainability System for LGU Mina (IWMESS) in the IT field by its requirement for a variety of specialized skills and technologies during the phases of development, implementation, as well as maintenance. Besides, the system development is a necessity because of the residents being able to report waste, being able to track rewards, and to access the information which implies the need of professionals with expertise in UI/UX design and cross-platform development. Such a strong web-based platform is required for LGU administrators to handle data, generate reports and monitor the entire system which creates the need for web developers and database administrators.

Cloud computing is considered as the factor providing the required scalability and accessibility while cybersecurity is the area where the experts are needed to secure the sensitive data and maintain the system's reliability. In summary, IWMESS not only stretches the limits of technology but also fosters the growth of the IT sector by making different IT skills in demand and by giving a push to the development of waste management and eco-friendly practices that are based on technology.

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EcoMina_Waste_Management_System/

```
|
|— assets/
|   |— css/
|   |   └─ style.css
|   |— js/
|   |   └─ script.js
|   |— images/
|   |   └─ logo.png
|   └─ uploads/
|       └─ recycling_proofs/
|
|— includes/
|   |— db.php
|   |— functions.php
|   |— header.php
|   |— footer.php
|   |— auth_check.php
|   └─ constants.php
|
|— admin/
|   |— admin.php
|   |— admin_dashboard.php
|   |— create_admin.php
|   |— manage_users.php
|   |— manage_categories.php
|   |— monitor_waste.php
|   |— monetization.php
|   └─ updates.php
```

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```
|   |— verify_recycling.php
page
|   |— rewards_admin.php
|   |— reports.php
|   |— logout.php
|
|— user/
|   |— homepage.php
|   |— dashboard.php
|   |— categorization.php
|   |— points.php
|   |— reward.php
|   |— claim_reward.php
|   |— monetization.php
|   |— updates.php
|   |— logout.php
|
|— auth/
|   |— index.php
|   |— register.php
|   |— login.php
|   |— login_admin.php
|   |— verify_email.php
|   |— logout.php
|
|— reports/
|   |— waste_summary.php
|   |— user_activity.php
|   |— reward_history.php |
```

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|— index.php
/auth/index.php
|— README.md
└— .htaccess

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Chapter V

Summary

The Integrated Waste Management and Environmental Sustainability System (IWMESS), also known as EcoMina, is our capstone project developed specifically for the Local Government Unit (LGU) of Mina, Iloilo. We created this system in response to the town's growing need for a more efficient, transparent, and community driven approach to waste management. Through our research, we observed that waste tracking and community participation were often limited due to manual processes and a lack of digital infrastructure. To address this, we designed EcoMina, a user-friendly web platform that allows residents to log recyclable waste, earn points, and redeem rewards such as mobile load credits. At the same time, it provides LGU administrators with real time analytics, reports, and monitoring tools to improve policy decisions and waste collection strategies.

We used PHP, JavaScript, and MySQL following the Software Development Life Cycle (SDLC) methodology to ensure the system was structured, secure, and scalable. We also conducted surveys, interviews, and field observations in Mina to understand current practices, challenges, and community readiness. From these insights, we built features that encourage public engagement—like an intuitive interface, gamified rewards, and community updates. Beyond being just a software project, EcoMina represents our shared vision of how technology can empower local communities to take part in environmental stewardship.

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Developing this system allowed us to combine our technical skills with a meaningful social goal helping Mina move closer to a cleaner, greener, and more sustainable future.

Conclusion

Our capstone project, the Integrated Waste Management and Environmental Sustainability System (IWMESS) or EcoMina, has been a meaningful journey that combined our passion for technology with our commitment to environmental sustainability. Through the development of this system, we aimed to provide a practical and innovative solution for the Local Government Unit (LGU) of Mina, Iloilo, addressing long standing issues in waste tracking, community engagement, and data driven decision making. By creating a platform where residents can easily log recyclable waste, earn and redeem points, and access educational resources, we were able to encourage active participation and environmental responsibility at the community level. On the other hand, the system's analytics and monitoring tools empower LGU officials to make informed and timely policy decisions.

This project taught us not only technical skills in web development, database design, and system architecture but also the importance of empathy, collaboration, and understanding the needs of the community we are serving. Seeing how technology can make a real difference in promoting sustainable practices inspired us to continue exploring how IT can be used for social good. EcoMina is more than just a system—it is our contribution to a cleaner, smarter, and more eco conscious Mina, and a reflection of our belief that small, tech-driven actions can create lasting positive change in society.

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Recommendation

Based on our experience in developing the Integrated Waste Management and Environmental Sustainability System (IWMESS) or EcoMina, we recommend that the Local Government Unit (LGU) of Mina continue enhancing and expanding the system's implementation beyond the pilot phase to fully realize its potential. It would be beneficial to integrate additional features such as a mobile application for easier access, especially for residents in remote areas, and possibly include IoT technologies like smart bins or sensors for real time waste monitoring. We also recommend that the LGU conduct continuous information drives and training sessions to educate residents about the importance of waste segregation and to guide them in using the platform effectively. Regular system maintenance, feedback collection, and updates should also be prioritized to ensure data accuracy, usability, and security.

From our perspective as developers, this project has shown us how vital collaboration between the LGU, community members, and technology developers is in achieving sustainability goals. We encourage future researchers and IT students to build upon this project perhaps by exploring mobile integration, AI based analytics, or expanded partnerships with recycling companies. Most importantly, we hope that EcoMina continues to inspire both residents and local leaders to embrace technology as a tool for environmental stewardship and community progress.

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References (APA Format)

Appendices:

Guerrero, L. A., Maas, G., & Hogland, W. (2013). Solid waste management challenges for cities in developing countries. *Waste Management*, 33(1), 220-232.
<https://doi.org/10.1016/j.wasman.2012.09.008>

Gonzales, M., & Magcale Macandog, D. (2021). Integrated solid waste management practices in Philippine municipalities: Challenges and prospects. *Journal of Environmental Science and Management*, 24(2), 56-68.

Magalang, A., & Caballes, M. (2019). Linking waste management and climate action in Philippine LGUs. *Philippine Journal of Environmental Sustainability*, 4(1), 45-59.

Mina Municipal Government. (2015). 10-Year Ecological Solid Waste Management Plan (2015-2024). Municipality of Mina, Iloilo.

Municipality of Mina. (2016). Comprehensive Land Use Plan, Volume 1 (2016-2025). Mina, Iloilo.

Philippine News Agency. (2018, May 10). Residents oppose proposed sanitary landfill in Iloilo town.
<https://www.pna.gov.ph/>

Republic Act No. 9003. (2000). Ecological Solid Waste Management Act of 2000. Official Gazette of the Republic of the Philippines. <https://www.officialgazette.gov.ph>

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Appendices

As part of our project, the appendices include several essential materials that support the design, development, and evaluation of the Integrated Waste Management and Environmental Sustainability System (IWMESS) or EcoMina. These appendices contain our survey questionnaires, which were distributed to residents, LGU personnel, and barangay officials to gather insights on existing waste management practices and their expectations for the new system. We also included system design diagrams such as the Entity Relationship Diagram (ERD), Data Flow Diagram (DFD), and Use Case Diagram, which visually represent the system's structure, data processes, and user interactions. Additionally, the appendices feature sample screenshots of the EcoMina website, showing the homepage, dashboard, reward system, and submission pages that demonstrate the platform's usability and functionality.

From a personal standpoint, compiling the appendices was one of the most rewarding parts of this project. It allowed us to reflect on how far we had come from conceptualization to actual system deployment. Including our raw survey data, observation notes, and testing results also helped us evaluate the real world impact of EcoMina on both the LGU and the community. These materials not only serve as supporting documents but also as tangible proof of our team's hard work, collaboration, and dedication to promoting environmental sustainability through technology.

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Appendix A: System Interface Screenshots

The screenshot displays the 'Recycling Monetization Report' for a user named Geran Salanid. The interface includes a header with the EcoMina logo and a 'Back to Home' link. The main content area is divided into several sections:

- Summary:** A light blue box with the greeting 'Hello, Geran Salanid! Here's your complete recycling monetization summary.' Below this, four cards show: Total Revenue (P1.00), Available Points (98), Total Recycled (107.1 kg), and Avg. Rate (P0.01).
- Environmental Impact:** Three cards showing CO₂ Saved (0.0 kg), Water Saved (0 L), and Energy Saved (0.0 kWh).
- Earnings by Recycling Category:** A table with columns: Category, Weight, Revenue, Points, and Avg./kg. It lists 'Glass Containers' with 1.0 kg weight, P1.00 revenue, 10 points, and P1.00/kg average.
- Recent Recycling Activities:** A table with columns: Date, Material, Weight, and Earnings. It shows a recent activity on Oct 29 for 'Glass Containers' (1.0 kg) earning P1.00.

Below the report, there is a 'Community Updates & Announcements' section with several posts:

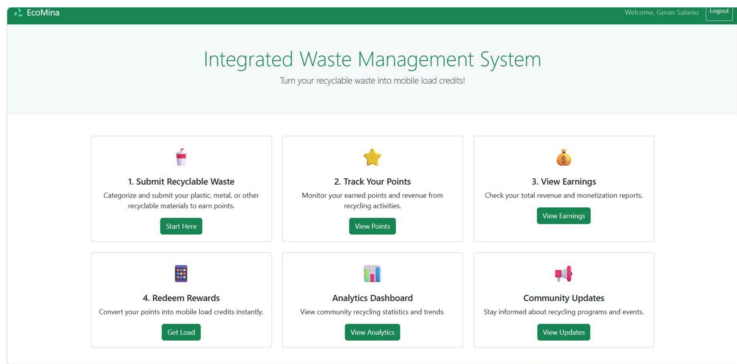
- New Reward System Launched!** (Posted: October 29, 2025 - 9:00 AM) - We're excited to announce our new mobile load rewards system! Convert your recycling points directly to mobile credits for all networks.
- Increased Rates for Metal Cans!** (Posted: October 26, 2025 - 2:30 PM) - Metal cans now earn **P22 per kg!** Aluminum and tin cans are among our highest-paying recyclable materials.
- Top Recycler of the Month** (Posted: October 22, 2025 - 10:15 AM) - Congratulations to our top recycler! Special bonuses awarded to users who recycled over 50kg this month.
- System Maintenance** (Posted: October 15, 2025 - 4:45 PM) - Monthly system maintenance scheduled for this weekend. The system will be unavailable for 2 hours on Sunday from 2:00 AM to 4:00 AM.
- Community Clean-up Drive** (Posted: October 8, 2025 - 8:30 AM) - Join our community clean-up drive this Saturday! Earn double points for all waste collected during the event.
- New Recycling Categories Added!** (Posted: October 24, 2025 - 11:00 AM) - We've optimized our recycling categories for better efficiency. Focus on plastic bottles (P18/kg) and metal cans (P22/kg) for the best return!

The screenshot displays the 'Mobile Load Rewards' section of the EcoMina interface. It includes a header with the EcoMina logo and a 'Back to Home' link. The main content area is divided into several sections:

- Welcome, Geran Salanid!** You currently have 98 points. Convert your eco-points to mobile load credit!
- Mobile Load Rewards:** Four cards showing different reward levels: P10 Mobile Load (100 points, Need 2 more points), P20 Mobile Load (200 points, Need 102 more points), P50 Mobile Load (500 points, Need 402 more points), and P100 Mobile Load (1000 points, Need 902 more points).
- How to Earn Points:** A section with six items: Plastic Bottles (15 points/kg, P18 revenue share), Metal Cans (12 points/kg, P22 revenue share), Paper & Cardboard (8 points/kg, P12 revenue share), Glass Containers (10 points/kg, P15 revenue share), Mixed Materials (Varies, Check category rates), and Regular Recycling (Earn points daily, Consistent recycling pays off).

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Recyclable Materials

Choose from our accepted recyclable materials. Each material earns different points based on its recycling value and environmental impact.

Plastic Bottles
PET bottles, containers
30 pts/kg P1/kg

Metal
Aluminum, tin, steel
100 pts/kg P10/kg
Copper, brass, iron
200 pts/kg P20/kg
Stainless steel
300 pts/kg P30/kg
[Learn More](#)

Paper & Cardboard
Boxes, newspapers
10 pts/kg P1/kg

Glass Containers
Bottles, jars
10 pts/kg P1/kg

Submit Recyclable Materials

Recycling Category *

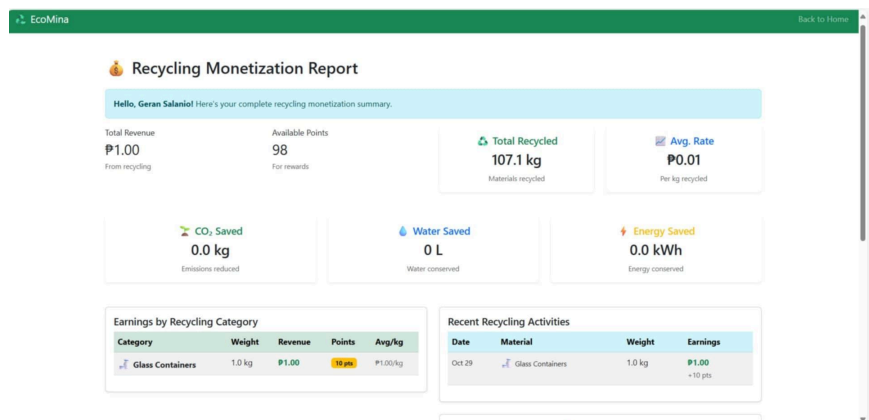
Choose what you're recycling...

Weight (kilograms) *

Enter weight in kg
Minimum: 0.1 kg

Proof of Recycling * REQUIRED

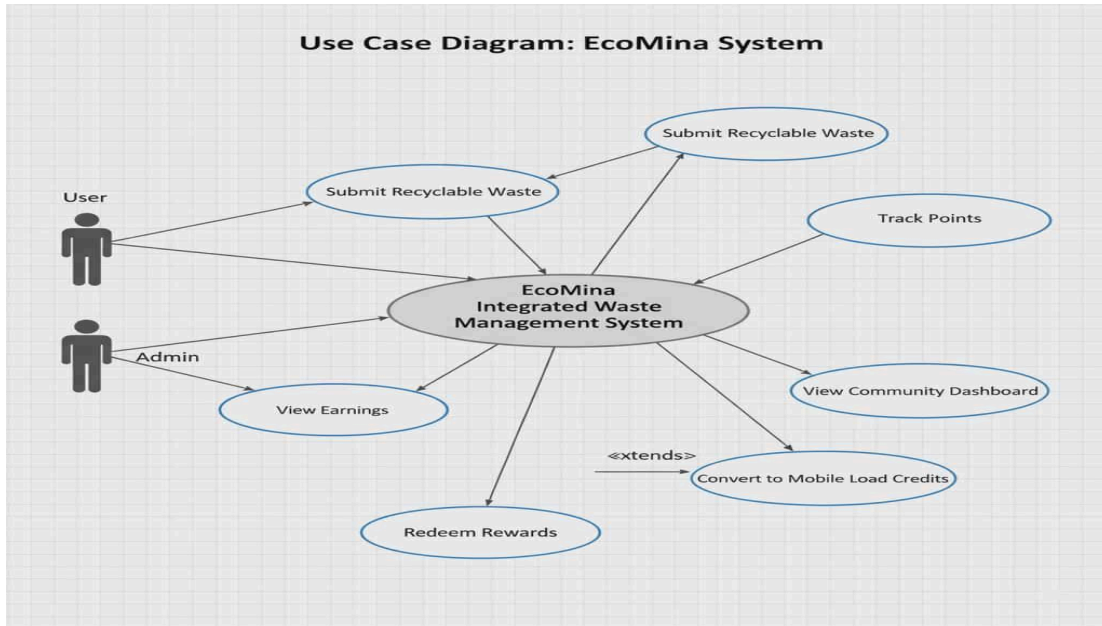
Upload Proof of Recycling
Take a clear photo showing the recyclable materials you're submitting
Accepted: JPG, PNG, GIF, WEBP (Max 5MB)



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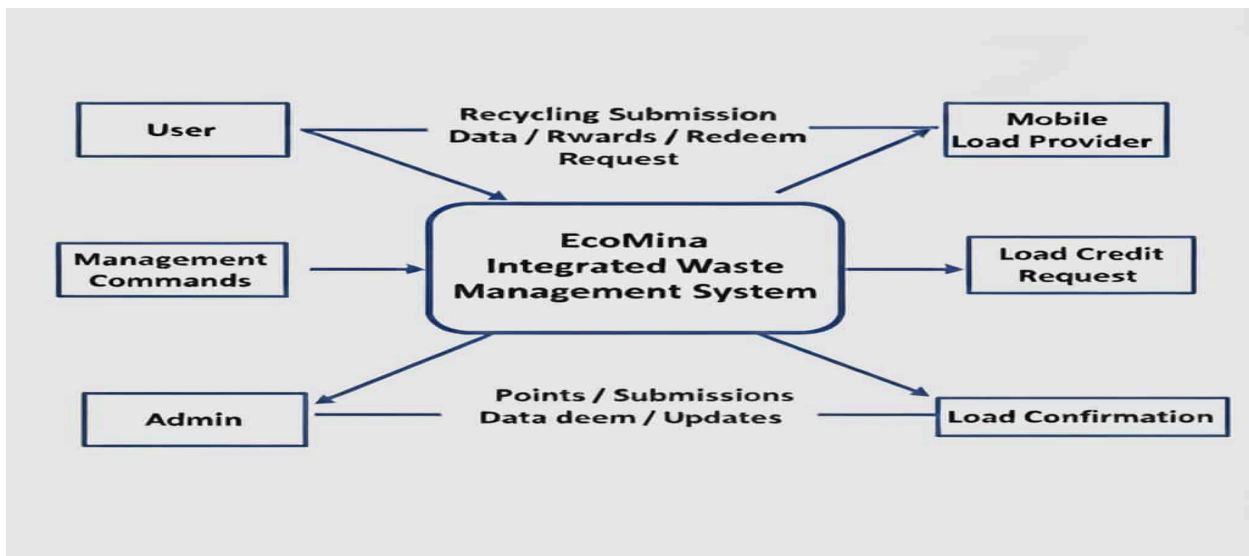
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Appendix B: Use Case Diagram



Appendix C: Data Flow Diagram

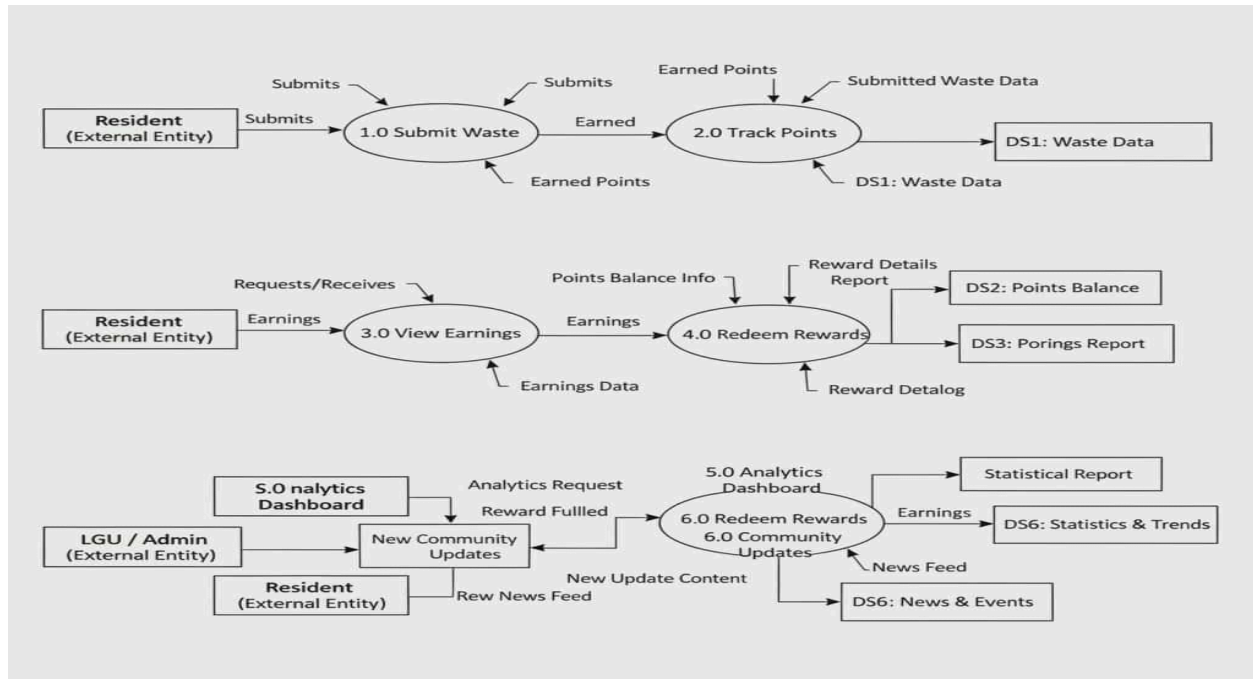
Level 0



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Level 1 Data Flow Diagram



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Appendix D: User Survey Form and Summary

Data Gather Techniques

Survey Questionnaire For System Assessment

The purpose of this survey was to obtain user feedback on "LGU MINA." The information gathered was used to determine how the system will be improved based on the user feedback.

Demographic Information

1. What is your role in waste management?

- LGU staff
- Barangay Officials
- Resident
- Other (Please specify): _____

2. How long have you been involved in community waste programs?

- Less than 1 year
- 1-3 years
- 4-6 years
- More than 6 years

3. What is your area or barangay? _____

Current Waste Management Practices

4. How is waste collection currently managed?

- Manual schedule (printed notice)
- Excel or spreadsheet tracking
- No formal system
- Other (Please specify): _____

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5. How satisfied are you with the current system?

- Very Satisfied
- Satisfied
- Neutral
- Unsatisfied
- Very Unsatisfied

6. What common problems do you face?

- Missing Collection
- Overflowing bins
- Lacks of update
- No awareness
- Other (Please specify): _____

Expectations for the Proposed System

7. What features would you find very useful for you?

- Community Updates
- Analytics Dashboards
- Submit Recycling Waste
- User accounts
- Redeem Rewards
- Other (Please specify): _____

8. How is recycling important?

- Very Important
- Important
- Neutral
- Not Important

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Concerns and Readiness for Adoption

9. What concerns do you have? (Select all that apply)

- Cost
- Internet Access
- Training Needs
- Resistance to change
- Other (Please specify): _____

10. How willing are you to use a new digital system?

- Very Willing
- Willing
- Neutral
- Unwilling
- Very Unwilling

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Appendix E: Sample Cover Page

PHINMA University of Iloilo

CITE Department

**Integrated Waste Management and Environmental
Sustainability System for LGU MINA**

A Capstone Project

Presented to the Faculty of the College of Information
Technology Education

In Partial Fulfillment of the Requirements
For the degree Bachelor of Science in Information Technology

By:

Flores, Sean Kristopher

Jumilla, Adrian

Salanio, Geran Lister

Valdez, Felice Jude

April 2026

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Appendix F: Grammarian CV
Character References

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Adrian Jumilla
Lanit Jaro, Iloilo
Iloilo, City 5000
adrianjumilla5@gmail.com
09318934065

October 30, 2025

Janine Montalban Tabares
OIC-MENRO
Cabatuan, Iloilo City
Iloilo City, 5000

Dear, Ms. Janine,

Subject: Proposal for Memorandum of Agreement to Use the
"Integrated Waste Management and Environmental
Sustainability System for LGU MINA"

We are happy to present our proposed Integrated Waste Management and Environmental Sustainability System for LGU MINA. This system is designed to gamify and manage recyclable waste submission by offering users digital incentives while providing LGU Mina with data analytics for program optimization and community engagement.

To make the use and setup of this system in your organization official, We suggest signing a Memorandum of Agreement (MOA).

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This agreement will outline the duties, terms, and conditions for both sides. It will ensure a clear understanding of the project goals, timelines, data safety, support, and maintenance.

The main features of your project are designed to digitize and incentivize waste management by creating a closed-loop system for both users and the local government unit (LGU).

- **User Incentivization:** Users are motivated to recycle through features like Submit Recyclable Waste, Track Your Points, View Earnings, and Redeem Rewards (converting points to mobile load or vouchers).
- **LGU Management & Optimization:** The LGU gains oversight and data through the Analytics Dashboard (for performance monitoring, trends, and revenue reports) and the Community Updates platform (for communication and engagement).

We are committed to supporting your team through the entire setup. process, including training and technical help to get the most out of the system.

We look forward to talking more about this proposal further and working together to improve your waste management system. Kindly let us know a good time to meet or if you require any additional information.

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Thank you for your consideration.

Sincerely,

Adrian Jumilla

UI UX/Documentator

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PHINMA UNIVERSITY OF ILOILO

College of Information and Technology Education

Rizal Street, Iloilo City Proper, Iloilo City 5000, Philippines

MEMORANDUM OF AGREEMENT (MOA)

Between Mina LGU And

Flores, Sean Kristopher

Jumilla, Adrian

Salanio, Geran Lister

Valdez, Felice Jude

Date: October 30, 2025

Purpose

This agreement is entered into by and between Janine Montalban Tabares and the above-named students from PHINMA University of Iloilo, to collaborate on the students' capstone project entitled "Integrated Waste Management and Environmental Sustainability System for LGU MINA".

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Scope of Agreement

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College of Information Technology Education

The Mina LGU agrees to provide support and resources, including:

- Agrees to give support and tools, like access to data, advice, and guidance.
- To help the organization set up the waste management system.
- Helping with planning and making sure everything works well.

The students agree to:

- Conduct research
- Develop and test the project
- Complete the capstone project according to the requirements

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Client Approval

By signing this document, MINA LGU acknowledges approval and support of the students' proposed capstone project. The department commits to timely cooperation and to providing relevant information and access as needed for the project's success.

Signatures:

For Janine Montalban Tabares:

JANINE MONTALBAN TABARES
MINA LGU
OIC-MENRO
Date: _____

For the Students:

GERAN LISTER SALANIO
Capstone Project Student
Date: _____

ADRIAN JUMILLA
Capstone Project Student
Date: _____

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COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

FLORES, SEAN KRISTOPHER
Capstone Project Student
Date: _____

FELICE JUDE VALDEZ
Capstone Project Student
Date: _____